

Dark Matter Failure-Regime Package

Specialist Companion

Status: Semantically Frozen Specialist Companion — Release-Ready

Public Role

This companion is the human-operable interface among the **scientific anchor**, the **Dark-Matter Failure-Regime Adapter / Contract**, the **Structural Flow Universal Kernel**, and the **concrete execution record produced under the scientific protocol**.

It explains how already-adjudicated scientific states are populated into the machine interface, how machine certificates are interpreted, which shortcuts are forbidden, and how a machine result is returned to the scientific claim language owned by the anchor.

It does **not** create a fourth scientific or machine authority.

Governing Scientific Anchor

Five Failure Regimes in Dark-Matter Light Probes: An Open-Source Pre-Result Demonstration Protocol

Use the governing accepted edition supplied with the release package.

Exact release identity belongs to the package manifest and to concrete execution records, not to this companion's semantic interface.

Machine Files

Dark-Matter Failure-Regime Adapter / Contract

One cumulative domain adapter / contract Lean file.

Structural Flow Universal Kernel

Separate domain-neutral Lean file supplied with the release package.

Purpose and Package Architecture

The Companion's Job

The companion exists so that an independent executor who did not build the package can determine, without hidden author interpretation:

- which scientific facts must already be established before Lean population;
- which machine field or burden receives each scientific fact;
- which machine route is appropriate at each stage of the experiment;
- what a successful certificate licenses;
- what an open, violated, blocked, or malformed route means;
- what the machine result does **not** establish;
- and which authority must be consulted when a scientific question remains unresolved.

The Four Public Layers

Scientific Anchor

The scientific anchor owns the experiment, physical interpretation, scientific-state meanings, freeze protocol, result taxonomy, claim ceiling, and empirical reporting rules.

Dark-Matter Failure-Regime Adapter / Contract

The adapter owns the machine representation of already-adjudicated scientific states and the conditional routing among the experiment-owned result states.

Structural Flow Universal Kernel

The Universal Kernel owns the reusable domain-neutral Structural Flow theorem machinery imported or invoked by the adapter.

Specialist Companion

The companion explains execution, population, interpretation, troubleshooting, compatibility, and reporting. It does not overrule the anchor, adapter, or kernel.

Separate-File Machine Architecture

The Universal Kernel and the Dark-Matter Failure-Regime Adapter / Contract are separate Lean files in the public package.

The adapter is checked against the compatible kernel. It is **not** merged into the kernel source as a new universal theorem family.

Primary Acceptance Criterion

The companion is complete only when an independent executor can determine the permitted next machine action from the scientific record and the released package without asking the author what the protocol “really meant.”

Authority Chain

Governing Chain

Physics / Domain Evidence supplies the physical realization, measurements, simulation state, model warrant, thresholds, and other domain-owned evidence.

The Governing Upstream Physics Authority remains the compatible Physics / Cosmology and dark-matter admission authority inherited by the scientific anchor. The anchor does not manufacture physical admission from Failure-Regime classification, and neither does this companion.

The Frozen Scientific Protocol adjudicates the experiment-owned scientific state and owns what that state means within the declared execution.

The Domain Adapter / Contract maps those already-adjudicated states into typed machine records, burdens, predicates, and certificates.

The Universal Kernel supplies reusable Structural Flow theorem machinery for the compatible universal seats used by the adapter.

Lean checks the encoded conditional structure.

The Companion explains how to move among those layers faithfully.

Authority Firewall

Lean does **not** decide whether:

- a dark-matter candidate is physically warranted;
- a mechanism is admitted;

- E occurred;
- P_0 , P, B, C, or A passed, failed, was not live, or remained unresolved;
- J_ROUTE passed or failed;
- a source or light prediction survived;
- O_2 is the strongest reasonable comparator;
- R_u exceeded ϵ_u under a scientifically valid calibration;
- a state-completeness ladder is scientifically adequate;
- K^* is physically real;
- an intervention is physically available;
- branch records are scientifically compatible;
- or an empirical claim is true of nature.

Those are scientific or domain adjudications. Lean checks what follows **if they are faithfully represented in the released contract**.

The companion also does not replace the anchor's inherited upstream authority. Physical-domain admission, mechanism warrant, and dark-matter identification burdens remain owned by the governing compatible physics authority named by the anchor. The experiment and machine package begin downstream of those admissions.

Closure Grammar

Scientifically valid, faithfully populated execution record

+

relevant machine certificate

→

anchor-licensed scientific result

Never:

machine certificate → empirical truth

Machine-clean is not physics-confirmed.

Package Identity and Compatibility

Stable Semantic Reference

The adapter and companion identify the scientific authority by stable title and by the rule:

Use the governing accepted edition supplied with the release package.

A harmless anchor-version bump does not by itself require a new adapter or companion build.

Compatibility Rule

Compatibility is semantic, not numerical.

A later anchor edition remains compatible only while machine-relevant meanings remain unchanged, including:

- condition and result meanings;
- prerequisite and liveness relations;
- burden ownership;
- route and branch routing;
- residual and reopening gates;
- and program-level licenses and blockers.

Copyediting, citation repair, layout changes, and explanatory expansion do not by themselves break compatibility.

Exact filenames, hashes, DOI or release identities, compiler environment, and execution bindings belong to the **release manifest** and the **concrete execution record**.

Vocabulary and Type Firewall

The scientific protocol and the Lean adapter use several vocabularies that may look similar while performing different jobs. They must not be collapsed into one another.

Universal Contract Vocabulary

Disposition.discharged

A declared contract burden has been positively satisfied under the supplied evidence relation.

Disposition.violated

A declared contract burden has been positively failed under the supplied evidence relation.

Disposition.open

The burden remains unresolved or otherwise not closed for the requested machine route.

These are **contract-status terms**. They are not automatically synonyms for experimental **PASS**, **FAIL**, or **UNRESOLVED**.

Route-Condition Vocabulary

ConditionStatus.pass

The registered scientific test for that live condition passed.

ConditionStatus.fail

The registered scientific test for that live condition failed.

ConditionStatus.notLive

The registered physical route does not require that condition.

NOT-LIVE is not **PASS** and is not **FAIL**.

ConditionStatus.unresolved

The condition is relevant, but the available intrinsic information cannot adjudicate it reliably enough for the requested classification.

UNRESOLVED is not **PASS** and is not **FAIL**.

J_ROUTE Vocabulary

JRouteStatus.pass

The registered physical route performed the task for which it was registered.

JRouteStatus.fail

The registered physical route did not perform that task under the frozen domain-native criterion.

JRouteStatus.unresolved

The registered criterion cannot adjudicate the route state reliably enough.

A J_ROUTE failure is necessary for a named route failure or structural-residual route, but **J_ROUTE FAIL alone does not identify a Failure Regime.**

Scale Vocabulary

ScaleRole.primary

The registered R_PRIMARY seat that owns the confirmatory first-break and closure result for that execution.

ScaleRole.secondary

A scientifically reportable diagnostic scale that cannot replace the registered primary result after target access.

E / produced

RouteRecord.produced is the machine representation of E: occurrence of the relevant interaction-produced difference.

Absence of E blocks named first-break evidence and blocks entry into the structural-residual pathway.

It is **not** silently converted into Dissolution or another named regime.

Branch Vocabulary

BranchBurden

The frozen pre-target branch-setup requirements required after the route/classification surface and before downstream branch adjudication.

ComparatorOutcome

The frozen O₂ comparison outcome after the branch survives measurement:

- sfWins;
- o2Wins;

- tie;
- or unresolved.

PrimaryResult

The branch-level terminal result licensed by the downstream chain:

- search failure;
- physicalization failure;
- source-bridge failure;
- light-bridge failure;
- measurement failure;
- U_1 fail;
- U_1 not earned;
- U_1 earned;
- or unresolved.

ClosureEffect

The branch-level foundational-closure consequence.

Layer-B terminal results do **not** themselves suspend or reopen foundational closure.

Residual and Reopening Vocabulary

ResidualBurden

The frozen protocol burdens governing observable novelty, SRW replication, state completeness, candidate audit, intervention availability, complete-result replication, and independent reproduction.

ResidualResult

The machine-result vocabulary for:

- novel-observable pressure;
- SRW candidate;
- replicated SRW;
- state-incompleteness resolution;

- closure suspension;
- FR₆ candidate;
- known-capacity reduction resolution;
- and FR₆ earned.

ResidualMilestoneCertificate

A diagnostic proof that one residual milestone has been reached.

Because residual predicates are cumulative, a milestone certificate is **not by itself a final scientific reporting license**.

ResidualReportingCertificate

The final or primary residual-reporting certificate.

It uses **PrimaryResidualResultEvidence**, which excludes stronger simultaneously active residual states so that the machine cannot under-report a suspension, restoration, or reopening as a weaker earlier milestone.

FoundationalClosureStatus

The machine-reporting status of the five-regime closure at the tested scope:

- **unchanged**;
- **suspended**;
- **restoredAtTestedScope**;
- or **reopenedAtTestedScope**.

restoredAtTestedScope

Closure had been placed under foundational pressure and suspended, but a known-capacity reduction or functional equivalence subsequently resolved the residual witness under the governing scientific adjudication.

This is **not** the same state as never having been pressured.

Program-Level Vocabulary

Program claim-support set

The exact branch set frozen before target access as the support set for one particular cross-branch or cross-regime claim.

Complete registered accounting

Every branch registered in the execution remains represented in the final program record, including adverse, unresolved, or untested branches.

Optional first-break and closure fields

`ProgramContract.firstBreakClass` and `ProgramContract.closureStatusAt` use `Option` values.

`none` is lawful where the scientific record is unresolved, untested, or otherwise has no earned value.

The operator must **not invent a class or closure state merely to complete a machine record**.

Closure pressure and finalization

`ProgramContract.closureTestPlacedUnderPressure` records whether the foundational closure test was actually placed under the relevant registered pressure.

`ProgramContract.closureStatusFinalized` records whether the closure result is scientifically final enough for program-level reporting.

Cross-regime U_1 license

A machine certificate that the frozen support set satisfies the encoded program-level U_1 aggregation burdens.

It does not replace branch records and does not adjudicate empirical truth.

Program closure-survival license

A separate machine certificate requiring:

- actual foundational pressure;
- finalized closure status;

- and a final status of **unchanged** or **restoredAtTestedScope** for every relevant frozen support branch.

Cross-regime U_1 success does **not** imply this license.

Critical Non-Equivalences

Disposition.discharged $\neq$ experimental PASS

Disposition.violated $\neq$ experimental FAIL, unless the specific mapping says so.

NOT-LIVE $\neq$ FAIL

UNRESOLVED $\neq$ FAIL

UNRESOLVED $\neq$ PASS

J_ROUTE FAIL $\neq$ named first break

PASS $\neq$ sixth Failure Regime

U_1 EARNED $\neq$ five-regime closure survived

CLOSURE SUSPENDED $\neq$ FR₆ EARNED

restoredAtTestedScope $\neq$ never-pressured **unchanged**

ResidualMilestoneCertificate $\neq$ final residual report

none $\neq$ invented first-break or closure value

machine certificate $\neq$ empirical adjudication

Scientific Record $\rightarrow$ Machine Population Ledger

Purpose

This ledger states **where machine values come from**.

It never supplies those values.

The governing rule is simple:

Scientific authority adjudicates the state.
The execution record preserves that adjudication.
The adapter receives the adjudicated state.
Lean checks what follows from it.

A machine field must therefore never be populated merely because a particular value makes a theorem easier to prove.

For each machine destination, the operator should be able to identify six things:

Machine destination — the exact adapter field or burden family.

Scientific owner / source — the anchor-owned record or domain adjudication that supplies the value.

Population time — when that value is fixed or adjudicated.

Forbidden source — information that may not lawfully be used to populate it.

Open / unresolved behavior — whether a non-final state is permitted and what that state means.

Claim consequence — what the populated value can constrain or license.

Route / Classification Population

RouteRecord.objectId

Scientific owner / source: the independently warranted physical candidate or execution-domain identity.

Population time: route registration, before confirmatory classification.

Forbidden source: the desired Failure-Regime label or held-out light morphology.

Open / unresolved: the identifier itself is not an adjudication state. If the physical object has not been identified, there is no valid route record to populate.

Claim consequence: binds the route record to its owning physical object without converting failure of one route into failure or dissolution of the entire object.

RouteRecord.routeId

Scientific owner / source: the minimum-admissible-route record.

Population time: pre-target route registration.

Forbidden source: a retrospective subdivision selected after target inspection because it produces a cleaner result.

Open / unresolved: there is no valid confirmatory route record without a pinned route identity.

Claim consequence: scopes J_ROUTE and the five condition tests to one declared physical participation relation.

RouteRecord.scaleId

Scientific owner / source: the registered physical-scale record.

Population time: pre-target freeze.

Forbidden source: a post-result scale selected because it yields a cleaner classification.

Open / unresolved: the scale identifier must be pinned for a confirmatory route.

Claim consequence: binds the result to the declared scale.

RouteRecord.scaleRole

Scientific owner / source: the R_PRIMARY declaration and any secondary-scale declarations.

Population time: pre-target freeze.

Forbidden source: post-result promotion of a secondary diagnostic scale.

Open / unresolved: there is no third machine state. Primary ownership must be explicit for a primary confirmatory certificate.

Claim consequence: only the primary scale can own

PrimaryConfirmatoryFirstBreak and tested-scope closure reporting.

A secondary scale may remain scientifically informative. It may not replace R_PRIMARY after the target is known.

RouteRecord.produced

Scientific owner / source: the frozen adjudication of E from permitted intrinsic or domain evidence.

Population time: route adjudication under the frozen classifier.

Forbidden source: held-out light morphology or the desired first-break class.

Open / unresolved: this machine field is a proposition. If E itself remains scientifically unresolved, the operator must not manufacture a completed confirmatory route by guessing its value.

Claim consequence: positive E is required for named first-break evidence and for entry into the Structural Residual Witness pathway.

Absence of E does not become Dissolution. It blocks the first-break and residual route.

RouteRecord.jRoute

Scientific owner / source: F_J(D_intrinsic) under the frozen J_ROUTE specification.

Population time: intrinsic route adjudication before the held-out probe is used as confirmatory evidence.

Forbidden source: the Failure-Regime label or held-out target statistic.

Open / unresolved: **JRouteStatus.unresolved** is explicit and blocks named confirmatory failure routing.

Claim consequence: J_ROUTE PASS blocks named first-break licensing. J_ROUTE FAIL is required for a named first break or an SRW, but is not sufficient by itself for either one.

RouteRecord.g.p0

Scientific owner / source: the frozen P₀ classifier.

Population time: intrinsic route adjudication.

Forbidden source: lensing morphology or any downstream branch result.

Open / unresolved: NOT-LIVE and UNRESOLVED remain distinct machine states.

Claim consequence: participates in prerequisite consistency and in discrimination among Dissolution, Shear, and every later first-break class.

RouteRecord.g.pl

Scientific owner / source: the frozen P classifier.

Population time: intrinsic route adjudication.

Forbidden source: morphology or a downstream failure.

Open / unresolved: NOT-LIVE and UNRESOLVED are preserved.

Claim consequence: participates in liveness closure, prerequisite consistency, and discrimination between Shear and later first-break classes.

RouteRecord.g.boundary

Scientific owner / source: the frozen Boundary test.

Population time: intrinsic route adjudication.

Forbidden source: a visible offset treated as a class label without an independently adjudicated route-level relation.

Open / unresolved: NOT-LIVE and UNRESOLVED are preserved.

Claim consequence: participates in discrimination between Drift and later first-break classes.

RouteRecord.g.cascade

Scientific owner / source: the frozen receiver-specific Cascade test.

Population time: intrinsic route adjudication.

Forbidden source: recurrence, repetition, or repeated morphology without a physically warranted receiving dependency.

Open / unresolved: NOT-LIVE and UNRESOLVED are preserved.

Claim consequence: participates in Echo / Lock discrimination.

RouteRecord.g.authorization

Scientific owner / source: the frozen Authorization or incorporation test.

Population time: intrinsic route adjudication.

Forbidden source: generic history dependence or hysteresis without the declared incorporation burden.

Open / unresolved: NOT-LIVE and UNRESOLVED are preserved.

Claim consequence: participates in Lock discrimination and in the all-five-no-break control state.

Route-Entry Burdens

Contract.entryState : EntryBurden → Disposition

This family receives the scientific adjudication of the route-entry requirements.

The covered burdens are:

- physicalCandidateWarrant;
- mechanismAdmission;
- participationRelationPinned;
- minimumAdmissibleRoute;
- primaryScaleOwnershipFrozen;
- jRouteCriterionFrozen;
- capacityTestsFrozen;
- intrinsicInputBoundaryPinned;
- heldOutProbeExcluded;
- and preTargetFreezeIntact.

Scientific owner / source: the pre-target execution and freeze record.

Population time: before confirmatory first-break machine licensing.

Forbidden source: a post-target rescue, an unsupported machine default, or an inference made solely because a later theorem is desired.

Open / unresolved: `Disposition.open` is explicit and blocks `EntryReady`.

Claim consequence: every required entry burden must be discharged before the adapter can license a confirmatory first-break route.

`Contract.entryEvidence` and `entryWarrant`

Scientific owner / source: the execution-owned evidence relation supporting the declared `entryState`.

Population time: with construction of the route contract.

Forbidden source: a bare assertion inserted solely to satisfy Lean.

Open / unresolved: the evidence must correspond to the actual declared disposition.

Claim consequence: prevents the contract status from functioning as an unbacked machine label.

The machine should therefore never contain:

“discharged because the proof needs discharged.”

It should contain:

“the scientific record adjudicated this burden as discharged, and this evidence witness maps that adjudication into the contract.”

Branch Adjudication Population

Once the route/classification surface is populated, the branch contract carries the experiment forward through source prediction, ordinary gravitational propagation, measurement, and the O₂ comparison.

Layer B does not repair Layer A. An unresolved or invalid route remains unresolved or invalid downstream.

`BranchContract.branchId`

Scientific owner / source: the registered branch identity.

Population time: branch registration.

Forbidden source: post-result regrouping of cases into a more favorable branch.

Open / unresolved: branch identity must be pinned for a branch record.

Claim consequence: scopes the branch-local result and later program accounting.

BranchContract.routeContract

Scientific owner / source: the already-populated route/classification contract for the same branch.

Population time: before branch-result adjudication.

Forbidden source: a route record reconstructed from the downstream outcome.

Open / unresolved: an open Layer-A state remains open. Layer B does not repair it.

Claim consequence: preserves route and classification authority upstream of branch outcomes.

BranchContract.declaredClass

Scientific owner / source: the pre-target branch registration identifying the first-break class the branch is constructed to instantiate.

Population time: before confirmatory target access.

Forbidden source: whichever class appears to fit the observed probe result best.

Open / unresolved: there is no machine UNRESOLVED constructor for this field. If a clean class cannot be scientifically declared, the branch must not be populated as though it were a clean declared-class branch.

Claim consequence: permits the adapter to distinguish confirmed physicalization from a mismatch between the registered branch and the intrinsic route classification.

Branch Freeze Burdens

BranchContract.branchState : BranchBurden → Disposition

The branch-state family carries the frozen setup requirements for downstream testing.

The relevant burdens are:

- `matchedControlFrozen`;
- `sourcePredictionFrozen`;
- `lightPredictionFrozen`;
- `measurementPipelineFrozen`;
- `ordinaryComparatorFrozen`;
- `utilityCriterionFrozen`;
- `targetHoldoutIntact`;
- and `branchFreezeVerified`.

Scientific owner / source: the pre-target freeze record.

Population time: freeze stage.

Forbidden source: a post-unblinding repair hidden inside the confirmatory record.

Open / unresolved: an open burden blocks `BranchSetupReady`.

Claim consequence: all required branch burdens must be discharged before the adapter can license downstream confirmatory branch results.

Search Adequacy

`BranchContract.searchState`

Scientific owner / source: the execution-owned adjudication of whether the registered domain actually placed the branch under adequate pressure.

Population time: execution or result stage under the frozen search rule.

Forbidden source: a downstream branch failure reinterpreted as evidence that the search itself was adequate.

Open / unresolved: `Disposition.open` is permitted and blocks adequate-result routing.

Claim consequence:

violated → **SEARCH FAILURE**

discharged → later branch results may proceed if all other required burdens hold.

Search failure means the execution failed to place the branch under its intended pressure. It is not positive evidence for or against the branch hypothesis.

Source Bridge

BranchContract.sourceBridgeState

Scientific owner / source: the anchor-owned adjudication of the frozen source prediction.

Population time: after clean physicalization of the registered branch.

Forbidden source: the held-out light result used to infer whether the source bridge succeeded.

Open / unresolved: an open state remains non-terminal. It may not be coerced into either failure or success.

Claim consequence:

a violated source bridge after confirmed physicalization → **SOURCE-BRIDGE FAILURE**

a discharged source bridge → required before later light and measurement stages can be licensed.

Light Bridge

BranchContract.lightBridgeState

Scientific owner / source: ordinary gravitational forward-model adjudication of the frozen probe consequence.

Population time: after the source prediction survives.

Forbidden source: measurement recovery substituted for the truth-level light bridge.

Open / unresolved: an open state blocks later terminal licenses.

Claim consequence:

a violated light bridge after source survival → **LIGHT-BRIDGE FAILURE**

a discharged light bridge → required before the measurement result can carry downstream.

Measurement

BranchContract.measurementState

Scientific owner / source: the frozen measurement and reconstruction recovery test.

Population time: after the truth-level light consequence has been established.

Forbidden source: absence of a source or truth-level light consequence recoded as a measurement failure.

Open / unresolved: an open state blocks O_2 / U_1 terminal results.

Claim consequence:

a violated measurement burden after earlier survival → **MEASUREMENT FAILURE**

a discharged measurement burden → required before comparator adjudication.

Ordinary-Physics Comparator

BranchContract.comparatorOutcome

Scientific owner / source: the frozen primary O_2 / U_1 comparison under the same target and the same pre-target information budget.

Population time: after branch survival through measurement.

Forbidden source: a secondary utility metric selected after seeing the target.

Open / unresolved: **ComparatorOutcome.unresolved** is explicit.

Claim consequence:

sfWins → U_1 **EARNED**, if all predecessor burdens survive.

o2Wins → **U₁ FAIL**.

tie → **U₁ NOT EARNED**.

unresolved → no stronger utility result is licensed.

U₁ EARNED is always scoped to the declared branch and primary utility criterion. It is not a foundational closure result.

Branch Result Certificate

ResultCertificate.primaryResult and **licensed**

Scientific owner / source: the terminal branch result derived from the populated **BranchContract** under the adapter predicates.

Population time: branch adjudication.

Forbidden source: a free-form global statement such as “the experiment passed” or “SF failed.”

Open / unresolved: **PrimaryResult.unresolved** is available only where the encoded unresolved route applies. Malformed or invalid inputs are not converted into UNRESOLVED merely to produce an output.

Claim consequence: the certificate reports the narrowest licensed branch result.

The branch layer preserves the location of failure:

search → **physicalization** → **source** → **light** → **measurement** → **O₂ / U₁**

A failure at one of those stages does not silently migrate upstream into foundational Failure-Regime pressure.

Layer-B results therefore remain branch-local unless and until the separately gated residual pathway is lawfully opened.

Route / First-Break Machine Interface

Operator Purpose

The route / first-break interface is the first machine seat at which the scientific execution can ask Lean for a named Failure-Regime certificate.

The machine does not:

- discover the route;
- choose the physical object;
- decide the primary scale;
- or infer condition states from a probe image.

Those facts must already exist in the frozen scientific record.

The operator's task is therefore **translation without reinterpretation**:

1. identify the already-warranted physical object, route, and scale;
2. populate the route record from the frozen intrinsic scientific adjudication;
3. populate and warrant the entry burdens;
4. preserve the four-state condition vocabulary exactly;
5. request only the machine proposition actually licensed by that record.

Build the Route Contract Before Asking for a Class

A valid route packet has two parts.

RouteRecord

This contains:

- object;
- route;
- scale;
- E / **produced**;
- J_ROUTE;
- and $G = (P_0, P, B, C, A)$.

Contract

This contains the **RouteRecord** plus the pre-outcome **EntryBurden** dispositions and their evidence / warrant relation.

EntryReady means every required **EntryBurden** is:

Disposition.discharged

An entry burden that is:

Disposition.open

or:

Disposition.violated

blocks the confirmatory first-break route.

The operator must return to the anchor-owned record rather than insert a favorable disposition merely to make a proof go through.

The entry surface is deliberately broader than the five condition tests. It includes:

- physical-candidate warrant;
- mechanism admission;
- participation-relation pinning;
- minimum-admissible-route status;
- frozen primary-scale ownership;
- frozen J_ROUTE;
- frozen capacity tests;
- the intrinsic-input boundary;
- held-out-probe exclusion;
- and intact pre-target freeze.

A clean G vector cannot compensate for a broken entry surface.

Preserve the Four-State G Vector

ConditionStatus.pass, **fail**, **notLive**, and **unresolved** are scientific states supplied by the frozen route adjudication.

They are not alternative spellings for Boolean true and false.

The operator must preserve three distinctions.

NOT-LIVE

The registered route does not require the condition.

UNRESOLVED

The condition is relevant, but it cannot be adjudicated to the required reliability.

FAIL

The live registered test positively failed.

RegistrationConsistent requires both scientific prerequisite consistency and prerequisite-closed liveness.

In practical terms:

- a downstream condition cannot remain live after a structurally required predecessor has been declared NOT-LIVE;
- and a later PASS cannot erase an earlier live FAIL.

An inconsistent record is not converted into the nearest regime.

It is an inadmissible confirmatory input until the scientific registration problem is repaired under the anchor's amendment rules.

J_ROUTE Is a Separate Gate

J_ROUTE answers whether the registered physical route performed its domain-native task.

It is not itself the first-break classifier.

The machine therefore distinguishes:

JRouteStatus.pass — the route succeeded under the frozen physical criterion.

JRouteStatus.fail — the route failed and may be eligible for a named first break or later residual analysis.

JRouteStatus.unresolved — the route-success criterion itself is not adjudicated.

LicensedFirstBreak requires:

J_ROUTE = FAIL

A route with J_ROUTE = PASS or UNRESOLVED cannot receive a named first-break license merely because one field of G has an attractive pattern.

LicensedFirstBreak

LicensedFirstBreak is the route-level machine proposition for a named class.

It requires all of the following:

- **EntryReady**;
- **J_ROUTE = FAIL**;
- **RegistrationConsistent G**;
- **FirstBreakEvidence** for the requested class.

FirstBreakEvidence is prefix-sensitive.

The requested break condition must FAIL, and every class-defining predecessor must PASS.

Later conditions do not redefine an earlier first break.

The five machine class patterns mirror the anchor's scientific decision matrix:

Dissolution — E present; P₀ FAIL.

Shear — E present; P₀ PASS; P FAIL.

Drift — E present; P₀ PASS; P PASS; B FAIL.

Echo — E present; P₀ PASS; P PASS; B PASS; C FAIL.

Lock — E present; P₀ PASS; P PASS; B PASS; C PASS; A FAIL.

NOT-LIVE or UNRESOLVED at a condition required for the requested class blocks that class.

A failed predecessor blocks every later class that presupposes its success.

PrimaryConfirmatoryFirstBreak

PrimaryConfirmatoryFirstBreak adds one burden to **LicensedFirstBreak**:

the record must own:

ScaleRole.primary

This distinction matters because the anchor permits scientifically useful secondary diagnostic scales while reserving the confirmatory first-break and closure result for the pre-registered R_PRIMARY seat.

A **FirstBreakCertificate** on a secondary scale can still encode a scoped route-level result.

But it is not a **PrimaryFirstBreakCertificate**, and it cannot replace the frozen primary result after target access.

PASS Control Is Not Depth 5 in Every Case

RoutePassControl is the domain-level non-failure control.

It requires:

- J_ROUTE = PASS;
- **RegistrationConsistent G**;
- no physically live foundational condition FAILS.

LicensedRoutePassControl adds the clean **EntryReady** surface.

This is intentionally broader than the kernel's fully-live depth-5 projection.

A route may be a valid PASS control while one or more downstream conditions are legitimately NOT-LIVE.

Such a record is not forced into the fully-live Boolean-depth interface.

AllFivePass and kernel depth 5 are used only where all five conditions are:

- fully live;
- resolved;
- and PASS.

The operator must not use depth 5 as a generic synonym for every scientific PASS control.

RouteFailureWithoutKnownBreak Is Only a Handoff Surface

RouteFailureWithoutKnownBreak means:

- J_ROUTE = FAIL;
- the registered G vector is consistent;
- no physically live known foundational condition FAILS.

This state is intentionally **not** named SRW.

It contains no claim that:

- the ordinary model library has been cleared;
- the mixture model has been cleared;
- the measurement pipeline has been cleared;
- the novelty threshold has been crossed;
- replication has succeeded;
- or the state-completeness ladder has been exhausted.

The correct operator interpretation is:

the route failed and the known live first-break tests did not classify that failure at this stage.

The next permitted machine action is a residual audit only after the relevant branch and ordinary-explanation gates are satisfied.

The operator must not report:

- SRW;
- closure suspension;
- or FR₆

from this route-level handoff alone.

KernelCompatible Is a Relation, Not a Coercion

The Universal Kernel represents foundational capacities propositionally.

The experiment represents P₀, P, B, C, and A using four scientific states.

The adapter therefore uses **KernelCompatible** to relate the two representations without erasing scientific information.

Under **ConditionCompatible**:

PASS constrains the corresponding kernel proposition to hold.

FAIL constrains the corresponding kernel proposition not to hold.

NOT-LIVE imposes no Boolean truth-value claim.

UNRESOLVED imposes no Boolean truth-value claim.

A kernel-compatible snapshot must therefore not be read backward as though Lean had adjudicated a NOT-LIVE or UNRESOLVED scientific condition.

For fully live, resolved records, **resolvedKernelProjection** supplies the narrower Boolean projection used for **firstBreakDepth**.

That projection is not the general route-record representation.

Named **FirstBreakEvidence** maps into one of the Universal Kernel's already-existing five **FailureClosure** regimes.

The adapter does not create a sixth kernel class.

It also does not replace the scientific route record with the kernel snapshot.

Route-Level Certificates

FirstBreakCertificate

Contains:

- the licensed **FirstBreakClass**;
- and a proof that **LicensedFirstBreak** holds for the populated **Contract**.

PrimaryFirstBreakCertificate

Contains:

- the licensed **FirstBreakClass**;
- and a proof that **PrimaryConfirmatoryFirstBreak** holds for the populated **Contract**.

Read either certificate in this order:

1. identify the **Contract** and therefore its **ObjectId** / **RouteId** / **ScaleId** binding;
2. identify the licensed class;
3. determine whether primary-scale ownership is part of the requested certificate;
4. return to the anchor for scientific meaning and reporting ceiling.

Neither certificate says that the entire **ObjectId** is a Dissolution, Shear, Drift, Echo, or Lock object.

The certificate remains **route-scoped**.

Schematic Invalid-Input Examples

Example A — Downstream Liveness Bypass

Suppose:

P₀ = PASS

P = NOT-LIVE

B = PASS

This is not a Drift-capable prefix.

B cannot remain live through a structurally required predecessor declared NOT-LIVE.

Registration consistency fails.

Do not relabel the record.

Example B — Unresolved Predecessor

Suppose:

P₀ = PASS

P = UNRESOLVED

B = FAIL

Do not report Drift.

The predecessor required to reach the Boundary break is unresolved.

Example C — Route Success Plus Apparent Condition Failure

Suppose:

J_ROUTE = PASS

P₀ = FAIL

Do not request a named first-break license.

The record requires upstream scientific adjudication, and J_ROUTE PASS blocks **LicensedFirstBreak**.

Example D — Secondary Diagnostic Class

Suppose a secondary scale cleanly satisfies the Shear pattern while the registered R_PRIMARY result is different or unresolved.

The secondary result remains scientifically reportable as a diagnostic.

It cannot be promoted to **PrimaryConfirmatoryFirstBreak** after target access.

Route-Interface Exit Rule

Proceed to branch adjudication only after the route / classification state required by the branch is scientifically meaningful and faithfully represented.

If:

- the route is malformed;
- entry remains open or violated;
- or intrinsic classification remains unresolved,

preserve that limitation.

Do not use later machine layers to repair an earlier scientific record.

Branch Adjudication Machine Interface

Operator Purpose

The branch interface begins after the route / classification seat.

Its job is to preserve the one-way experimental chain from:

search

→ **physicalization**

→ **source**

- **light**
- **measurement**
- **O₂ / U₁ comparison**

The branch layer does not decide whether the five-regime architecture is foundationally complete.

It localizes lower-level outcomes so that failure at one scientific link is not promoted into a stronger upstream claim.

Build **BranchContract** From the Existing Route Contract

BranchContract carries:

- **branchId**;
- the already-populated route contract;
- **declaredClass** — the class the branch was built to instantiate;
- pre-target **BranchBurden** dispositions;
- **searchState**;
- **sourceBridgeState**;
- **lightBridgeState**;
- **measurementState**;
- **comparatorOutcome**.

The route contract is not reconstructed from downstream evidence.

The branch must inherit the same scientific route record that was registered before the target result was known.

BranchSetupReady requires every required **BranchBurden** to be discharged:

- matched control;
- source prediction;
- light prediction;
- measurement pipeline;
- O₂ comparator;
- utility criterion;
- target holdout;

- branch freeze verification.

An open or violated branch-setup burden blocks downstream branch-result licensing.

Missing setup must not be reinterpreted as a negative scientific result.

Search Comes First

SearchAdequate, **SearchFailed**, and **SearchOpen** are separate from physicalization.

SEARCH FAILURE means the registered domain did not place the branch under the intended pressure.

It is terminal for that execution path before the branch's scientific prediction is tested.

Search failure is not:

- physicalization failure;
- or evidence that the branch prediction is false.

Conversely, **SearchAdequate** is required before downstream branch-result routes can proceed.

Physicalization Adjudication

PhysicalizationAdjudicationReady requires:

- route **EntryReady**;
- primary-scale ownership;
- **RegistrationConsistent G**.

It says the route surface is clean enough for the declared branch class to be judged.

It does **not** say the declared class has been confirmed.

DeclaredPhysicalizationConfirmed

The route has a **PrimaryConfirmatoryFirstBreak** in the branch's **declaredClass**.

KnownPhysicalizationMismatch

This is a resolved scientific mismatch, not merely absence of confirmation.

It can arise when:

- E is absent;
- the route is a valid PASS control;
- or a different known primary first break is licensed.

IntrinsicClassificationOpen

This is different.

It preserves a scientifically unresolved route state instead of converting it into a mismatch.

An unresolved J_ROUTE or unresolved intrinsic failure state does not become **PHYSICALIZATION FAILURE** merely because the declared class was not proved.

Malformed registration and violated entry burdens remain invalid upstream inputs.

They are not converted into scientific UNRESOLVED by the branch layer.

RouteUnknownHandoff

RouteUnknownHandoff requires:

- clean branch setup;
- adequate search;
- clean physicalization-adjudication entry;
- E present;
- and RouteFailureWithoutKnownBreak.

This is the branch-to-residual handoff.

It is still **not an SRW**.

The operator should read it as:

the branch has reached a genuine primary route failure not classified by the known live first-break map; the residual layer may now test whether ordinary explanations and the residual/newness gates are satisfied.

Do not report foundational pressure from `RouteUnknownHandoff` alone.

One-Way Terminal Result Order

The branch chain is:

SEARCH

→ PHYSICALIZATION

→ SOURCE BRIDGE

→ LIGHT BRIDGE

→ MEASUREMENT

→ O_2 / U_1

A later terminal result presupposes survival of every required earlier link.

The operator reports the **earliest live failed link**, not the most dramatic downstream description.

PrimaryResult States

SEARCH FAILURE

Licensed when `BranchSetupReady` holds and `searchState` is violated.

Scientific meaning: the registered search / target surface failed to place the branch under the intended pressure.

Foundational consequence: untested by that branch.

PHYSICALIZATION FAILURE

Licensed only after adequate search and a resolved

`KnownPhysicalizationMismatch`.

Scientific meaning: the proposed ordinary-physics realization did not instantiate the registered class.

Foundational consequence: unchanged.

SOURCE-BRIDGE FAILURE

Licensed only after adequate search and **DeclaredPhysicalizationConfirmed**, when **sourceBridgeState** is violated.

Scientific meaning: a cleanly classified route did not produce the frozen source consequence.

Foundational consequence: unchanged.

LIGHT-BRIDGE FAILURE

Licensed only after source survival, when the ordinary-gravity / probe translation is violated.

Scientific meaning: the source difference survived but did not produce the registered truth-level light consequence.

Foundational consequence: unchanged.

MEASUREMENT FAILURE

Licensed only after source and light survival, when **measurementState** is violated.

Scientific meaning: the truth-level probe consequence existed, but the frozen measurement / reconstruction path did not recover it.

Foundational consequence: unchanged.

U₁ FAIL

Licensed only after survival through measurement and:

comparatorOutcome = o2Wins

Scientific meaning: O₂ outperformed the SF-organized prediction on the frozen primary utility criterion.

Foundational consequence: unchanged.

U₁ NOT EARNED

Licensed only after survival through measurement and:

comparatorOutcome = tie

Scientific meaning: the branch may still support derivational convergence, but incremental SF utility is not earned.

Foundational consequence: unchanged.

U₁ EARNED

Licensed only after survival through measurement and:

comparatorOutcome = sfWins

Scientific meaning: branch- and metric-scoped incremental predictive utility under the anchor's frozen comparison rule.

Foundational consequence: unchanged.

U₁ EARNED is not a closure-survival certificate.

UNRESOLVED

Licensed only at the earliest live branch gate whose predecessors have survived.

It is not a generic fallback for malformed input or a previously failed earlier stage.

Scientific meaning: the branch cannot yet receive one of the resolved branch-level terminal results under the populated record.

Foundational consequence: no result.

ResultCertificate

ResultCertificate contains:

- one PrimaryResult;

- and a proof that **ResultEvidence** licenses that result for the populated **BranchContract**.

The operator must read a **ResultCertificate** together with:

- branch identity;
- inherited route / object / scale identity;
- the exact predecessor-survival chain;
- **closureEffect** for that **PrimaryResult**;
- and the anchor's scientific result taxonomy.

The certificate does not independently prove that the source, light, measurement, or comparator adjudications were scientifically correct.

It proves only that the encoded terminal result follows from the supplied machine record.

ClosureEffect Firewall

The branch layer intentionally has only three closure effects:

- **unchanged**;
- **untestedByBranch**;
- **noResult**.

No branch-level terminal result suspends or reopens the five-regime closure.

This remains true even for:

U₁ EARNED

The firewall prevents two opposite mistakes:

- a lower-level failure being exaggerated into foundational novelty;
- a positive utility result being exaggerated into proof that foundational closure survived.

Foundational pressure is adjudicated separately in the residual layer.

Adapter-Coverage Boundary

PrimaryResult is selective rather than exhaustive. DCR, purity, known mixture, coalescence, underpowered states, and other execution-schema states remain anchor-owned unless they feed an encoded machine route. Absence of a dedicated constructor does not invalidate those scientific states; the global Machine-Coverage Boundary below governs when an omitted distinction creates adapter pressure. Novel-observable, SRW, closure-suspension, and FR₆ states remain in the residual layer rather than the branch-result layer.

Branch-Interface Exit Rule

If the branch terminates at:

- SEARCH;
- PHYSICALIZATION;
- SOURCE;
- LIGHT;
- MEASUREMENT;
- or O₂ / U₁,

report that narrowest licensed result and stop the branch chain there.

Enter the residual layer only when:

- the scientific record has opened the residual pathway;
- and the route state satisfies the residual handoff burdens.

Do not use the residual layer to rescue an ordinary branch failure.

Residual / Reopening Population

The residual layer is where the protocol becomes deliberately difficult to escalate.

Its purpose is not to make novelty easy to recognize. Its purpose is to ensure that an unusual observable cannot become foundational pressure until ordinary explanations, route status, replication, state completeness, candidate newness, reduction, intervention, irreducibility, and reproduction have all been handled in the order established by the scientific anchor.

ResidualContract.branch

Scientific owner / source: the already-populated **BranchContract** for the same execution branch.

Population time: before residual adjudication.

Forbidden source: branch facts reconstructed retrospectively from the residual conclusion.

Open / unresolved: any unresolved or invalid upstream branch state remains a blocker.

Claim consequence: preserves the lower-level scientific chain before foundational escalation.

The SRW route inherits **RouteUnknownHandoff** from this branch. It does not rebuild a weaker route-failure test inside the residual layer.

Residual-Protocol Freeze Burdens

ResidualContract.residualState : ResidualBurden → Disposition

This family carries the frozen residual and newness decision machinery.

The relevant burdens include:

- **observableLibraryFrozen;**
- **noveltyThresholdFrozen;**
- **nonFoundationalRecheckRuleFrozen;**
- **srwReplicationRuleFrozen;**
- **stateCompletenessLadderFrozenPreTarget;**
- **candidateAuditRuleFrozen;**
- **interventionAvailabilityRuleFrozen;**
- **completeResultReplicationRuleFrozen;**
- and **independentReproductionRuleFrozen.**

Scientific owner / source: the pre-frozen residual and foundational-newness protocol records.

Population time: at the corresponding pre-target or pre-candidate freeze seats required by the anchor.

Forbidden source: rules invented after an anomaly is observed in order to promote it.

Open / unresolved: an open burden blocks **ResidualEntryReady**.

Claim consequence: no residual or newness result may outrun the frozen audit machinery that governs it.

Observable Novelty

ResidualContract.ruAboveThreshold

Scientific owner / source: the frozen R_u / ϵ_u adjudication after application of the closed observable library.

Population time: novelty evaluation.

Forbidden source: visual unfamiliarity alone.

Open / unresolved: unresolved novelty calibration must not be populated as **True**.

Claim consequence: **True** is necessary for **NOVEL-OBSERVABLE PRESSURE**, but it is insufficient for an SRW.

An unusual-looking target is therefore not enough.

The machine question is whether the registered closed library leaves a residual satisfying:

$$R_u > \epsilon_u$$

under the scientifically valid, pre-frozen calibration.

Non-Foundational Recheck

ResidualContract.nonFoundationalExplanationsCleared

Scientific owner / source: the registered recheck of measurement, source modeling, mixture, nuisance structure, numerical effects, and other non-foundational explanations.

Population time: after observable novelty pressure and before SRW licensing.

Forbidden source: the assumption that an unexplained morphology is therefore structural novelty.

Open / unresolved: unresolved ordinary explanations block escalation.

Claim consequence: required before route-level residual pressure can be licensed.

Structural Residual Witness Entry

SRWCandidate / RouteUnknownHandoff

Scientific owner / source: the already-populated **BranchContract**, together with the residual gates above.

Population time: only after observable novelty pressure has been established and non-foundational explanations have been cleared.

Forbidden source: rebuilding a weaker route-failure predicate manually.

Open / unresolved: all of the following block SRW entry:

- search failure;
- search remaining open;
- a secondary rather than primary scale;
- broken branch setup;
- open or violated route-entry burdens;
- absent E;
- unresolved required route conditions;
- or an already-known first break.

Claim consequence: the released adapter requires the full **RouteUnknownHandoff**, preserving primary-scale ownership and the one-way branch-to-residual handoff.

An SRW is therefore not:

“something strange remained.”

It is much narrower:

a genuine primary-scale route failure, with E present, after a valid upstream branch handoff, for which no live known first break accounts for the failure and ordinary non-foundational explanations have been cleared under the frozen protocol.

SRW Replication

ResidualContract.srwReplicationSucceeded

Scientific owner / source: the predeclared SRW replication execution.

Population time: after an SRW candidate has been established.

Forbidden source: repeated analysis of the same uncontrolled witness counted as independent replication.

Open / unresolved: unresolved or failed replication does not suspend closure.

Claim consequence: replication is required for **ReplicatedSRW** and every later foundational state.

One SRW candidate does not reopen or suspend anything.

State-Completeness Audit

ResidualContract.stateCompletenessExhausted

ResidualContract.richerOrdinaryStateResolves

ResidualContract.maximalStateResidualPersists

Scientific owner / source: the completed $S_0 \dots S$ state-completeness audit under the pre-frozen ladder.

Population time: after a replicated SRW.

Forbidden source: ad hoc state enrichment after the anomaly or preference for novelty over an ordinary explanation.

Open / unresolved: unresolved exhaustion, resolution, or persistence does not license closure suspension.

Claim consequence:

richerOrdinaryStateResolves = True supports
STATE-INCOMPLETENESS RESOLVED.

Otherwise, a replicated SRW together with:

- exhaustion of the pre-frozen state ladder;
- no richer ordinary-state resolution;
- and persistence of the residual at maximal state

supports a **MAXIMAL-STATE STRUCTURAL RESIDUAL** and therefore **CLOSURE SUSPENDED.**

The central anti-novelty question here is:

Did the apparent new structural failure disappear when the route was represented with a richer ordinary physical state already admitted by the frozen protocol?

If yes, the novelty route stops.

Positive Candidate Proposal

ResidualContract.candidateProposed

Scientific owner / source: a positive K^* proposal or a proposed new irreducible loss surface after closure suspension.

Population time: only after the maximal-state structural residual has been established.

Forbidden source: attaching a new label to the original anomalous morphology.

Open / unresolved: absence of a candidate leaves closure suspended while the residual persists.

Claim consequence: a positive candidate is required for **FR₆ CANDIDATE.**

The anomaly itself is not K^* .

K^* must identify a proposed missing structural job or irreducible loss surface.

Candidate-Newness Disposition

ResidualContract.candidateNewnessState : Disposition

Scientific owner / source: the scientific adjudication of the candidate-newness audit as OPEN, VIOLATED, or DISCHARGED.

Population time: after **FR6Candidate** is opened and while the positive newness audit is resolved.

Forbidden source: machine convenience used to convert an incomplete audit into PASS or FAIL.

Open / unresolved:

Disposition.open keeps the candidate under audit.

When no stronger result has been earned, this supports **FR₆ CANDIDATE** as the final residual report.

Claim consequence:

Disposition.violated means the proposed candidate failed its newness burden.

That does **not** automatically mean the underlying residual disappeared.

Closure either remains suspended or is restored only if a known-capacity reduction actually resolves the witness.

Disposition.discharged is necessary for **FR₆ EARNED**, but it is not sufficient by itself.

Positive Newness Burdens

The adapter carries the following candidate burdens:

- **ResidualContract.distinctPhysicalJob;**
- **ResidualContract.physicalCarrierIdentified;**
- **ResidualContract.passTestAvailable;**
- **ResidualContract.failTestAvailable;**

- `ResidualContract.predecessorRelationsPinned;`
- `ResidualContract.downstreamCapabilityDepends;`
- and `ResidualContract.necessityEstablished.`

Scientific owner / source: the foundational-newness audit owned by the anchor.

Population time: after FR₆ candidate entry.

Forbidden source: semantic renaming, descriptive novelty, or correlation alone.

Open / unresolved: an incomplete burden blocks FR₆ EARNED.

Claim consequence: these fields form positive components of `NewnessAuditComplete.`

A new capacity is not earned because it has a new name.

It must do a distinct job.

It must have a physical carrier.

It must have PASS and FAIL tests.

Its predecessor relations must be identifiable.

A downstream capability must genuinely depend on it.

And its necessity must be established rather than merely correlated with successful routes.

Intervention Burden

The intervention surface uses:

- `ResidualContract.interventionAvailabilityAdjudicated;`
- `ResidualContract.interventionAvailable;`
- `ResidualContract.removalTestPassed;`
- and `ResidualContract.restorationTestPassed.`

Scientific owner / source: the predeclared intervention-availability rule and physical intervention evidence where intervention is available.

Population time: foundational-newness audit.

Forbidden source: a post-hoc claim of unavailability used as a waiver, or a default PASS assigned to an intervention that was never performed.

Open / unresolved: unadjudicated availability blocks FR₆ EARNED.

Where intervention is physically available, both removal and restoration tests must pass.

Claim consequence: these fields encode the conditional intervention burden.

Unavailability does not count as failure.

It also does not erase the other newness burdens.

Reduction to Known Structure

ResidualContract.reducesToKnownFive

ResidualContract.functionallyEquivalentToKnown

Scientific owner / source: the reduction audit against P₀, P, B, C, A and against ordinary functional redescription.

Population time: foundational-newness audit.

Forbidden source: treating a new name or surface-form difference as evidence of structural novelty.

Open / unresolved: unresolved reduction or functional equivalence must not be populated as **False** simply because **False** allows the candidate to advance.

Claim consequence: either field being **True** blocks FR₆ EARNED.

A positive reduction may also participate in closure restoration—but only through the separate witness-resolution adjudication described below.

Does the Known Reduction Actually Resolve the Witness?

ResidualContract.knownCapacityReductionResolvesWitness

Scientific owner / source: the domain-owned scientific adjudication that the identified known-capacity reduction or functional equivalence actually resolves the maximal-state residual witness.

Population time: after a candidate is found to reduce to known structure or functional equivalence.

Forbidden source: assuming that rejection of the candidate automatically restores closure.

Open / unresolved: if the candidate fails but the witness remains unresolved, closure remains suspended.

Claim consequence: together with:

- `FR6Candidate;`
- `candidateNewnessState = Disposition.violated;`
- and established reduction to or equivalence with known structure,

a `True` value licenses:

KnownCapacityReductionResolved

and the closure status:

restoredAtTestedScope

This distinction is load-bearing.

Candidate rejected $\neq$ witness resolved.

The known explanation must actually account for the residual that created the suspension.

Irreducibility, Replication, and Independent Reproduction

The final positive newness burdens include:

- `ResidualContract.uniqueRemovalOrIrreducible;`
- `ResidualContract.completeResultReplicated;`
- and `ResidualContract.independentlyReproduced.`

Scientific owner / source: the irreducibility audit, complete-result replication, and independent procedural reproduction under the frozen rules.

Population time: the final positive newness audit.

Forbidden source: the original witness alone, or a continuous severity difference presented as a new irreducible loss surface.

Open / unresolved: failure or incompleteness blocks FR₆ EARNED.

Claim consequence: these are final positive components of **NewnessAuditComplete**.

Milestone Certificates

ResidualMilestoneCertificate.milestone / licensed

Scientific owner / source: a diagnostic request over **ResidualResultEvidence**.

Population time: during residual and newness adjudication.

Forbidden source: using a milestone certificate as the primary final report when a stronger state is active.

Open / unresolved: the certificate is unavailable if the requested milestone itself has not been established.

Claim consequence: proves only that the milestone was reached.

It does **not** settle the strongest supported residual state.

This matters because the residual predicates are cumulative.

A record that eventually reaches FR₆ EARNED has necessarily passed through earlier milestones.

Those earlier truths remain diagnostically useful.

They do not remain valid as the **primary final report**.

Final Residual Reporting Certificate

ResidualReportingCertificate.result / licensed

Scientific owner / source: the final or primary residual result proved through **PrimaryResidualResultEvidence**.

Population time: when the residual record is ready for scientific reporting.

Forbidden source: free selection of a weaker cumulative milestone.

Open / unresolved: the final certificate is unavailable until one strongest-supported constructor is lawfully satisfied.

Claim consequence: **reportingClosureStatus** maps the final residual result to:

- **unchanged;**
- **suspended;**
- **restoredAtTestedScope;**
- or **reopenedAtTestedScope**.

Residual / Reopening Machine Interface

Operator Purpose

The residual interface is the experiment's escalation firewall.

Its job is to prevent the transition:

unusual target → new foundational regime

from skipping the ordinary explanatory and scientific burdens between those two claims.

The full route is deliberately longer:

observable novelty

→ **ordinary-explanation recheck**

→ **primary route handoff**

→ **SRW candidate**

→ **replication**

→ **state-completeness audit**

→ **maximal-state residual**

- **closure suspension**
- *positive K^* candidate*
- **newness / reduction / intervention / irreducibility audit**
- **restoration or FR_6**

A known branch result, ordinary nuisance explanation, known mixture, measurement limitation, search failure, invalid physicalization, secondary-scale route, or known first break cannot become foundational novelty merely because it is passed into the residual namespace.

ResidualContract and ResidualEntryReady

ResidualContract carries the populated **BranchContract** plus the residual and foundational-newness state.

ResidualEntryReady requires every required **ResidualBurden** to be discharged.

Those burdens establish that the novelty and newness decision machinery was frozen before it was used to promote an anomaly.

An open or violated residual burden blocks the stronger route that depends upon it.

The operator must not turn an incomplete protocol into evidence for novelty.

NovelObservablePressure

NovelObservablePressure requires:

- **ResidualEntryReady**;
- and **ruAboveThreshold**.

Its scientific reading is narrow:

the frozen observable / model library left $R_u > \epsilon_u$ under the supplied scientific adjudication.

It does not establish:

- a route failure;
- E ;

- exclusion of known first breaks;
- an SRW;
- closure suspension;
- or FR₆.

Scientific states such as **NOVELTY DISCRIMINATION UNDERPOWERED** remain anchor-owned execution states rather than dedicated residual-result constructors.

If the novelty calibration did not validly authorize the R_u / ϵ_u comparison, do not populate the machine novelty gate as though it had.

SRWCandidate — Strict Branch Handoff

SRWCandidate requires all three:

- **NovelObservablePressure**;
- **nonFoundationalExplanationsCleared**;
- and **RouteUnknownHandoff** on the inherited **BranchContract**.

This is deliberately stronger than merely requiring J_ROUTE failure with no known first break.

RouteUnknownHandoff carries the full branch-to-residual history:

- branch setup is ready;
- search is adequate;
- physicalization adjudication is ready;
- Layer-A entry and registration are valid;
- the route owns the primary scale;
- E is present;
- J_ROUTE fails;
- and no live known first break accounts for the route failure.

Therefore none of the following can lawfully create an SRW candidate:

- a secondary-scale route;
- SEARCH FAILURE;
- search remaining open;

- broken branch freeze or setup;
- open or violated physical/domain entry;
- absent E;
- unresolved required condition;
- or a known first break.

SRW is not a generic receptacle for whatever the lower branch could not classify.

ReplicatedSRW

ReplicatedSRW requires:

SRWCandidate + successful predeclared replication

One witness does not suspend closure.

StateIncompletenessResolved

A replicated SRW may disappear when the same route is represented through a richer ordinary physical state from the pre-frozen state-completeness ladder.

When that richer state resolves the witness:

StateIncompletenessResolved

is the appropriate machine result.

This is an ordinary-state resolution.

It is not foundational failure.

MaximalStateStructuralResidual

A maximal-state structural residual requires:

- replicated SRW;
- exhaustion of the pre-frozen state-completeness ladder;
- no richer admitted ordinary state resolving the witness;

- and persistence of the residual at the maximally warranted state.

If a richer ordinary state resolves the witness, the operator must stop there.

The residual may not continue toward suspension merely because the earlier anomaly was interesting.

ClosureSuspended

ClosureSuspended is the maximal-state structural residual state at R_PRIMARY.

Suspension is an active reporting restriction.

While suspension remains unresolved:

the five-regime closure may not be reported as having survived at the tested scope.

Suspension is also not FR₆ EARNED.

The correct state may remain:

CLOSURE SUSPENDED

for as long as the witness remains real and unresolved.

FR6Candidate

FR6Candidate requires:

- **ClosureSuspended**;
- and a positive candidate proposal.

The positive candidate may be:

- a proposed new capacity K*;
- or a proposed new irreducible loss surface.

The candidate's later status is carried separately by **candidateNewnessState**.

Candidate-Newness Outcomes

Candidate remains **OPEN**

`candidateNewnessState = Disposition.open`

The candidate remains under newness adjudication.

When no stronger result is earned, **FR₆ CANDIDATE** remains the final residual report.

Candidate is **VIOLATED**

`candidateNewnessState = Disposition.violated`

The proposed candidate has failed its newness burden.

That does not answer whether the original maximal-state residual has disappeared.

Two paths remain possible:

candidate rejected + witness unresolved → CLOSURE SUSPENDED

or:

**candidate rejected + known reduction actually resolves witness →
restoredAtTestedScope**

Candidate is **DISCHARGED**

`candidateNewnessState = Disposition.discharged`

The candidate-newness disposition has been positively satisfied.

This is required for **FR₆ EARNED**, but the complete positive newness audit must also close.

Known-Capacity Reduction and Restoration

A failed candidate restores closure only when all of the following are true:

- **FR₆ CANDIDATE** had been opened;
- `candidateNewnessState = Disposition.violated`;

- the candidate reduces to the known five or is functionally equivalent to known structure;
- and the domain-owned scientific adjudication establishes that the reduction actually resolves the residual witness.

That final requirement is carried by:

knownCapacityReductionResolvesWitness

If the candidate reduces but the witness remains unexplained, closure remains suspended.

If the reduction resolves the witness, the result becomes:

KnownCapacityReductionResolved

and the closure status becomes:

restoredAtTestedScope

This status preserves the scientific history:

closure was genuinely pressured → closure was suspended → the residual was later absorbed by known structure.

It is not equivalent to a route whose closure was never pressured.

Intervention Availability

InterventionBurdenSatisfied first requires intervention availability to have been scientifically adjudicated.

If intervention is physically available, both:

- removal;
- and restoration

must pass under the frozen rule.

If intervention is physically unavailable, the machine does not invent a failed intervention.

But unavailability does not waive the remaining newness burdens.

NewnessAuditComplete

The positive newness audit carries the candidate burdens for:

- a distinct physical job;
- an identified physical carrier;
- available PASS and FAIL tests;
- pinned predecessor relations;
- a downstream capability that depends upon the candidate;
- necessity;
- intervention where physically available;
- non-reduction to P_0 , P , B , C , or A ;
- functional non-equivalence to known structure;
- unique-removal or irreducibility;
- complete-result replication;
- and independent procedural reproduction.

A positive candidate must survive the whole burden.

FR6Earned

FR6Earned requires:

- FR6Candidate;
- `candidateNewnessState = Disposition.discharged;`
- and NewnessAuditComplete.

Its scientific meaning remains bounded:

the tested architecture has reopened as incomplete at the demonstrated scope because a necessary irreducible first-break distinction survived the complete newness path.

It does not:

- establish a sixth observable morphology;
- silently generalize beyond the execution;
- rewrite the entire dark-matter domain;

- or add a sixth constructor to the Universal Kernel.

The kernel remains the current five-regime architecture until universal structure is separately adjudicated.

Milestones Versus Final Reporting

Residual predicates are cumulative.

A record that has reached closure suspension has also passed through earlier novelty and SRW states.

For audit and debugging, the adapter therefore supplies:

ResidualMilestoneCertificate

A milestone certificate means:

 this stage was reached.

It does not necessarily mean:

 this is the strongest state that should now be reported.

The final reporting surface is:

ResidualReportingCertificate

using:

PrimaryResidualResultEvidence

The stronger final-report predicates exclude weaker cumulative states whenever a stronger state is simultaneously active.

This prevents a suspended, restored, or reopened architecture from being reported merely as an earlier SRW or novelty milestone.

Final Residual Constructors

The final reporting constructors behave as follows.

novelObservablePressure

Observable novelty exists, but SRW has not been established.

srwCandidate

An SRW candidate exists, but replication has not been established.

replicatedSRW

The SRW has replicated, but neither ordinary-state resolution nor closure suspension has been established.

stateIncompletenessResolved

A richer ordinary state resolved the witness.

closureSuspended

A replicated maximal-state structural residual persists, and no stronger restoration or reopening result has been earned.

fr6Candidate

A positive candidate exists and remains open under the newness audit.

knownCapacityReductionResolved

A known-capacity reduction or functional equivalence actually resolved the residual witness.

fr6Earned

The complete positive newness burden has been discharged.

Final Closure-Status Map

reportingClosureStatus maps the final residual report as follows:

NOVEL-OBSERVABLE PRESSURE → **unchanged**

SRW CANDIDATE → **unchanged**

REPLICATED SRW → unchanged

STATE-INCOMPLETENESS RESOLVED → unchanged

CLOSURE SUSPENDED → suspended

FR₆ CANDIDATE → suspended

KNOWN-CAPACITY REDUCTION RESOLVED → restoredAtTestedScope

FR₆ EARNED → reopenedAtTestedScope

ClosureSurvivalReportAllowed is therefore:

allowed for unchanged

allowed for restoredAtTestedScope

not allowed for suspended

not allowed for reopenedAtTestedScope

The two allowed histories remain scientifically different.

unchanged means the tested pressure never produced an unresolved suspension or reopening.

restoredAtTestedScope means foundational pressure was real, suspension occurred, and the witness was subsequently resolved under known structure.

That historical distinction must remain visible in program reporting.

Residual-Interface Exit Rule

At the end of residual adjudication, record:

- the strongest available ResidualReportingCertificate;
- any diagnostic milestone certificates needed for audit, clearly labeled as milestones;
- the mapped foundational closure status;
- the scientific evidence and frozen execution identity supporting every populated premise;

- and the narrowest anchor-owned scientific statement licensed by the final state.

If no final residual certificate is available, do **not** choose a convenient earlier result merely to complete the record.

Instead, locate the actual obstruction:

- open burden;
- violated burden;
- malformed record;
- unresolved scientific adjudication;
- or incorrect machine route.

Then return to the authority that owns that obstruction.

Program-Level Population

Program-level reporting begins only after branch-level records exist.

The program surface does not create a new scientific result by averaging branches together. Its job is to preserve:

- the complete registered branch universe;
- the exact support set frozen for one aggregate claim;
- branch-local result identity;
- first-break identity where actually earned;
- closure history where actually finalized;
- compatibility among supporting branches;
- any admitted cross-scale relation;
- and the distinction between predictive utility and foundational closure.

The program contract therefore carries both **accounting burdens** and **claim-license burdens**.

Program Object and Claim Content

ProgramContract.programObject

ProgramContract.claimContent

Scientific owner / source: the pre-target declaration of the program-level object and the exact shared content of the intended aggregate claim.

Population time: program-claim freeze.

Forbidden source: post-result reformulation broad enough to absorb whatever branches happened to succeed.

Open / unresolved: both must be declared before **ProgramClaimFreezeReady** can hold.

Claim consequence: together they define the exact aggregate scientific statement the machine is being asked to support.

The machine is therefore not asked:

Did the program succeed?

It is asked something narrower:

Does this exact predeclared support set satisfy the encoded burdens for this exact predeclared program object and claim content?

Complete Registered Branch Universe

ProgramContract.registeredBranch

Scientific owner / source: the complete branch universe registered in the execution freeze.

Population time: pre-target execution registration.

Forbidden source: the final list of successful branches.

Open / unresolved: the registered universe itself must be known before complete accounting can be established.

Claim consequence: every registered branch remains visible in final program reporting.

A registered branch does not disappear because it:

- was never tested;
- remained unresolved;

- returned an adverse result;
- failed to earn U_1 ;
- or fell outside one later aggregate claim.

Registration determines the execution universe.

It does not guarantee success.

Frozen Aggregate Support Set

ProgramContract.claimBranch

Scientific owner / source: the exact branch support set frozen for one particular aggregate claim.

Population time: pre-target program-claim freeze.

Forbidden source: selection of successful branches after target access.

Open / unresolved: the support set must be frozen before an aggregate certificate can be requested.

Claim consequence: defines which branch results must jointly satisfy the burdens for that exact claim.

This creates two different branch sets:

registered branches — everything the execution committed to carrying;

claim branches — the exact predeclared subset intended to support one particular aggregate statement.

A branch can be registered without belonging to a particular aggregate support set.

But a claim branch must belong to the registered execution universe.

Final Branch Accounting

ProgramContract.recordPresent

ProgramContract.tested

ProgramContract.u1Earned

Scientific owner / source: the final execution inventory, branch-execution status, and branch-local U_1 adjudication.

Population time: program reporting after branch results are available.

Forbidden source:

- branch registration treated as though it were completed testing;
- or a program-level average used to upgrade a branch that did not earn U_1 locally.

Open / unresolved:

A missing record blocks **CompleteRegisteredAccounting**.

An untested support branch blocks an aggregate claim that requires it to have been tested.

A support branch that did not earn U_1 blocks an aggregate U_1 license that requires U_1 from every support branch.

Claim consequence: preserves complete accounting and the branch-local character of the evidence.

Program aggregation cannot create branch evidence that the branch itself did not earn.

First-Break Class at Program Level

ProgramContract.firstBreakClass : BranchId → Option FirstBreakClass

Scientific owner / source: the branch's licensed primary first-break class, where one exists.

Population time: branch-result record.

Forbidden source:

- morphology;
- aggregate inference;
- or an invented class assigned merely because program-level comparison needs one.

Open / unresolved: use:

none

for an unresolved, untested, or otherwise no-result branch.

Claim consequence: structural cross-regime support requires actual class values.

StructurallyCrossRegime therefore requires at least two supporting branches with:

some k_1

and:

some k_2

such that:

$k_1 \neq k_2$

A branch with **firstBreakClass** = **none** cannot be silently used as one side of the cross-regime distinction.

Closure Status at Program Level

ProgramContract.closureStatusAt : BranchId → Option FoundationalClosureStatus

Scientific owner / source: the final branch or scope closure status derived from the strongest residual report where such a closure result exists.

Population time: final branch/result record.

Forbidden source:

- U_1 success treated as closure survival;
- absence of a suspension proof treated as **unchanged**;
- or an invented closure status assigned to an untested or unresolved branch.

Open / unresolved: use:

none

where no earned final closure status exists.

Claim consequence:

- **none** blocks program closure-survival licensing;
- **suspended** blocks it;
- **reopenedAtTestedScope** blocks it;
- **unchanged** may support it if the additional pressure and finalization burdens are satisfied;
- **restoredAtTestedScope** may also support it if those burdens are satisfied.

The use of **Option** is deliberate.

No result is a valid machine state.

The operator must not invent a positive state simply to make the program record total.

Was Closure Actually Put Under Pressure?

ProgramContract.closureTestPlacedUnderPressure

Scientific owner / source: the scientific record showing that the five-regime closure test was actually exposed to the relevant registered foundational pressure.

Population time: final branch or result record.

Forbidden source: ordinary branch testing or ordinary branch success treated as though foundational closure had been challenged.

Open / unresolved: **False** blocks **ProgramClosureSurvivalLicensed**.

Claim consequence: prevents a closure surface that was never tested from being reported as having survived.

This distinction is essential.

A branch can succeed through:

- source prediction;
- light prediction;
- measurement;
- and U_1 comparison

without ever placing foundational closure under residual pressure.

That branch may support a utility result.

It does not thereby establish that the five-regime architecture survived a foundational challenge.

Is the Closure Status Final Enough to Aggregate?

ProgramContract.closureStatusFinalized

Scientific owner / source: the scientific adjudication that the branch's closure status is final enough for the requested program-level report.

Population time: after the relevant residual and newness pathway is complete for that reporting purpose.

Forbidden source: a temporary intermediate state treated as final because the program needs a reportable value.

Open / unresolved: **False** blocks **ProgramClosureSurvivalLicensed**.

Claim consequence: prevents a provisional closure state from supporting a final survival claim.

A machine-populated closure state therefore needs two independent reporting warrants:

the closure test was genuinely placed under pressure

and:

the resulting closure status is finalized for this reporting purpose

Program-Object and Compatibility Checks

The program contract also carries:

- `ProgramContract.matchesDeclaredProgramObject;`
- `ProgramContract.matchesDeclaredClaimContent;`
- `ProgramContract.mutuallyCompatible;`
- and `ProgramContract.scaleRelationAdmitted.`

Scientific owner / source: domain-owned final checks under the pre-frozen program object, claim content, compatibility rule, and cross-scale rule.

Population time: program reporting.

Forbidden source: post-result redefinition, harmonization, or retrospective compatibility.

Open / unresolved: any mismatch, incompatibility, or unadmitted scale relation blocks the aggregate claim.

Claim consequence: preserves one exact program object and one exact shared claim content across the support set.

Different branch results may all be scientifically valid while still being incompatible with one another for purposes of one aggregate claim.

Program aggregation is therefore not a vote-counting exercise.

Program-Claim Freeze Burdens

The following fields record whether the intended aggregate claim was frozen before target access:

- `ProgramContract.programObjectDeclaredPreTarget;`
- `ProgramContract.claimContentDeclaredPreTarget;`
- `ProgramContract.claimBranchSetFrozenPreTarget;`
- `ProgramContract.compatibilityRuleFrozenPreTarget;`
- and `ProgramContract.scaleRelationRuleFrozenPreTarget.`

Scientific owner / source: the freeze bundle for the intended program claim.

Population time: before target access.

Forbidden source: a reconstructed “freeze” assembled after the outcomes are known.

Open / unresolved: any missing pre-target burden blocks

ProgramClaimFreezeReady.

Claim consequence: together these fields create the program-level anti-cherry-picking surface.

The aggregate claim therefore has to exist in recognizable form **before** the data reveal which branch combination would be most favorable.

Program-Level Aggregation and Reporting

Operator Purpose

Program aggregation must preserve:

- branch identity;
- frozen claim scope;
- structural class identity where earned;
- scale relations;
- complete accounting;
- and final closure history.

Its job is **not** to create one global score for the experiment.

Four distinct questions remain separate:

1. What happened in every registered branch?
2. Which branches actually earned the states needed by a particular aggregate claim?
3. Does the exact frozen aggregate U_1 claim satisfy its machine burdens?
4. Was foundational closure actually pressured and finally adjudicated strongly enough to support a bounded survival statement?

Registered Branches and Claim Branches

ProgramContract contains two different branch predicates.

registeredBranch

This is the complete execution universe.

claimBranch

This is the exact pre-target support set for one particular aggregate claim.

A branch may be registered without belonging to the support set for a particular claim.

But every claim branch must come from the registered universe.

No registered branch may disappear from final accounting merely because it was:

- adverse;
- unresolved;
- untested;
- or inconvenient to the final narrative.

ProgramClaimFreezeReady

ProgramClaimFreezeReady packages five pre-target commitments:

- the program object was declared;
- the exact claim content was declared;
- the claim-support branch set was frozen;
- the compatibility rule was frozen;
- and the cross-scale relation rule was frozen.

This is the machine-facing expression of the program's anti-cherry-picking commitment.

A later aggregate claim cannot lawfully redefine:

- what the program object was;
- what common content the branches were supposed to support;
- which branches counted;
- what “compatible” meant;
- or how distinct R_PRIMARY scales were allowed to relate.

CompleteRegisteredAccounting

CompleteRegisteredAccounting requires a final record for every registered branch.

This includes branches that:

- were never ultimately tested;
- remained unresolved;
- returned adverse results;
- did not earn U_1 ;
- or did not belong to the support set for one aggregate claim.

Complete accounting does not mean complete success.

It means **nothing registered is silently erased**.

Cross-Regime U_1 Support

CrossRegimeUtilityClaimLicensed

The cross-regime utility certificate requires:

- **ProgramClaimFreezeReady;**
- complete registered accounting;
- the support set to lie within the registered universe;
- every support branch to have been tested;
- every support branch to have earned branch-scoped U_1 ;
- every support branch to match the declared program object;
- every support branch to match the exact shared claim content;
- mutual compatibility among the support branches;
- an admitted relation among any distinct $R_PRIMARY$ scales;
- and at least two support branches with actual, earned, distinct first-break classes.

The final requirement means the machine needs real class values:

some k_1

and:

some k_2

with:

$k_1 \neq k_2$

A branch whose `firstBreakClass = none` cannot be used to manufacture cross-regime diversity.

What the Cross-Regime U_1 Certificate Means

`CrossRegimeUtilityClaimLicensed` means:

the exact pre-frozen support set satisfies the encoded program-level burdens for the exact declared aggregate U_1 claim.

It does **not** mean:

- every registered branch succeeded;
- every Failure Regime was tested;
- the five-regime architecture is foundationally complete;
- closure survived;
- or the program achieved some unscoped global validation.

The certificate is bounded by:

`programObject + claimContent + claimBranch`

Program Closure-Survival License

`ProgramClosureSurvivalLicensed`

Program-level closure survival is a separate machine question.

It requires:

- `ProgramClaimFreezeReady`;
- complete registered accounting;
- the support set to lie within the registered universe;
- every support branch to have been tested;
- the foundational closure test to have actually been placed under pressure for every support branch;
- the resulting closure status to have been finalized for every support branch;
- and every support branch to have a final closure status of either:

some unchanged

or:

some restoredAtTestedScope

The certificate is blocked by:

none

some suspended

or:

some reopenedAtTestedScope

Why Actual Pressure Is Required

The burden:

closureTestPlacedUnderPressure

is load-bearing.

A branch whose foundational closure was never challenged cannot support the statement:

closure survived the test.

Even if every ordinary branch prediction succeeded, foundational survival remains untested unless the residual pathway actually placed the five-regime architecture under the relevant pressure.

The machine therefore refuses to equate:

ordinary success

with:

foundational survival

Why Finalization Is Required

The burden:

closureStatusFinalized

is equally important.

A temporary closure state may occur during an active residual or newness investigation.

Program reporting must not freeze that temporary state into a final aggregate statement merely because aggregation is being performed at that moment.

A final program survival claim therefore requires both:

pressure occurred

and:

the resulting closure status is final enough for the requested report

Two Lawful Survival Histories

The program certificate permits two final statuses.

unchanged

The architecture was placed under the relevant tested pressure and did not enter an unresolved suspension or reopening state.

restoredAtTestedScope

The architecture was genuinely pressured and suspended, but a later known-capacity reduction or functional equivalence actually resolved the witness.

Both may support a bounded survival report.

But their histories are different.

The program record must preserve that difference.

A restored architecture should not be rewritten as though it had never been pressured.

Utility and Closure Are Separate Licenses

A program may possess:

CrossRegimeUtilityClaimLicensed

while lacking:

ProgramClosureSurvivalLicensed

The reverse can also occur.

The two certificates answer different scientific questions.

U₁ question

Did the SF-organized prediction add incremental predictive utility against O₂ under the frozen criterion?

Closure question

After actual foundational pressure and finalized adjudication, did the five-regime architecture remain intact or become restored rather than remaining suspended or reopening at the tested scope?

Neither certificate substitutes for the other.

In particular:

Cross-regime U₁ success $\neq$ closure survival

and:

closure survival $\neq$ incremental U₁

Anti-Cherry-Picking Example

Suppose an execution registers three schematic branches:

A, B, C

Before target access, one aggregate utility claim freezes the support set:

{A, B}

After execution:

- A earns U_1 ;
- B does not earn U_1 ;
- C earns U_1 .

The final record must retain all three branches.

A and C remain scientifically reportable at branch level.

But C cannot replace B inside the pre-frozen aggregate support set merely because C succeeded.

The intended aggregate claim:

{A, B}

is unavailable.

A later program may prospectively register a new claim using a different support set.

It may not rewrite the earlier freeze.

Closure Example — No Foundational Pressure

Suppose Branch A was tested through:

- source prediction;
- light prediction;
- measurement;
- and utility comparison.

But its foundational closure surface was never actually placed under residual pressure.

Then:

closureTestPlacedUnderPressure A = False

ProgramClosureSurvivalLicensed is blocked.

This remains true even if the record were mistakenly populated with:

closureStatusAt A = some unchanged

The proper response is to repair the scientific/machine record.

Do not infer survival from the absence of foundational pressure.

Closure Example — Restored

Suppose Branch B reached:

CLOSURE SUSPENDED

under genuine foundational pressure.

A later known-capacity reduction then actually resolved the witness.

Its final closure state is:

some restoredAtTestedScope

If:

- the closure test was genuinely under pressure;
- the status is finalized;
- and the remaining program burdens are satisfied,

that branch may support the bounded program survival certificate.

The record must still preserve that suspension occurred before restoration.

Closure Example — Unresolved

Suppose Branch C has no finalized closure result.

Then either:

closureStatusAt C = none

or:

closureStatusFinalized C = False

or both.

Branch C cannot support a final program closure-survival statement.

The machine does not convert unresolved closure into **unchanged**.

Program-Interface Exit Rule

Program reporting should preserve four distinct outputs.

Complete branch accounting

What happened in every registered branch?

Earned branch identities

Which branches have earned first-break and closure values, and which properly remain **none**?

Aggregate utility license

Is the exact pre-frozen aggregate U_1 claim licensed for its declared program object, content, and support set?

Foundational closure license

Was closure actually placed under pressure, finally adjudicated, and licensed as **unchanged** or **restoredAtTestedScope** across the requested support surface?

These questions must not be collapsed into one global:

PASS

or:

FAIL

A program can contain multiple scientifically meaningful outcomes simultaneously.

The machine architecture preserves their locations rather than averaging them away.

Certificate Interpretation

General Rule

A machine certificate is a **typed proof carrier for one encoded conditional consequence**.

It must always be interpreted together with:

- the exact populated execution record;
- the scientific evidence that supplied that record;
- the scope and identity of the route, branch, residual, or program object;
- and the claim ceiling owned by the scientific anchor.

A machine-clean certificate does not validate incorrectly populated empirical premises.

For every certificate, the operator should ask four questions.

What exact record is this certificate bound to?

What proposition does it prove?

Is the certificate diagnostic, or is it the final reportable machine state?

What scientific statement does the anchor license from that proposition?

The certificate never answers the fourth question by itself.

Certificate Ceilings

FirstBreakCertificate

This certificate licenses **one named route-level first break**.

It does not establish primary-scale ownership unless that burden is separately present.

It does not say that the entire physical object “is” Dissolution, Shear, Drift, Echo, or Lock.

Its scope remains the registered route.

PrimaryFirstBreakCertificate

This certificate licenses the named route-level first break at the registered **primary confirmatory scale**.

It therefore includes the R_PRIMARY ownership required for the confirmatory first-break result.

It still remains route-scoped rather than whole-object scoped.

ResultCertificate

This certificate licenses one ordered Layer-B branch result.

Its result belongs to the chain:

SEARCH

- **PHYSICALIZATION**
- **SOURCE BRIDGE**
- **LIGHT BRIDGE**
- **MEASUREMENT**
- **O₂ / U₁**

No Layer-B **ResultCertificate** suspends or reopens foundational closure.

That remains true even when the result is U₁ **EARNED**.

ResidualMilestoneCertificate

This is a **diagnostic certificate**.

It proves that one residual milestone was reached.

Because the residual pathway is cumulative, earlier milestones may remain true after a stronger state has been reached.

A milestone certificate must therefore never be used to under-report a stronger active final state.

For example:

SRW reached

may remain historically true after:

CLOSURE SUSPENDED

has become the correct final state.

The milestone remains useful for audit.

It no longer owns the primary report.

ResidualReportingCertificate

This is the final residual / foundational-newness reporting surface.

Read its **result** together with:

reportingClosureStatus

The final mapped closure state can be:

unchanged

suspended

restoredAtTestedScope

or:

reopenedAtTestedScope

This certificate is specifically designed to prevent a stronger active residual state from being hidden behind an earlier cumulative milestone.

CrossRegimeUtilityClaimLicensed

This certificate licenses only the exact frozen aggregate U_1 claim identified by:

programObject + claimContent + claimBranch

It does not establish a general program success claim.

It does not establish foundational closure survival.

It does not permit substitution of successful branches after target access.

ProgramClosureSurvivalLicensed

This certificate licenses only the bounded closure-survival report for the frozen support surface.

It requires:

actual foundational pressure

plus:

finalized closure adjudication

plus final allowed closure states across the relevant support set.

It preserves the difference between:

unchanged

and:

restoredAtTestedScope

A restored result is not rewritten as though suspension never occurred.

Absence of a Certificate

The absence of a requested certificate has **no single scientific meaning**.

A missing proof may indicate:

- an open scientific burden;
- a violated burden;
- malformed or inconsistent population;
- a requested claim stronger than the record supports;
- use of the wrong machine route;
- a scientific state that is intentionally anchor-owned rather than represented as a Lean conclusion;
- or a formal engineering defect.

The operator must diagnose which case applies before drawing any scientific conclusion.

No proof does not automatically mean **scientific failure**.

Likewise, inability to obtain one strong certificate may coexist with a narrower valid scientific result.

Machine-Coverage Boundary

The adapter intentionally does not duplicate every anchor-owned scientific state as a theorem constructor.

Examples of states that may remain primarily in the scientific execution record include:

- DCR provenance;
- known-mixture states;
- purity states;
- coalescence states;
- underpowered calibration states;
- some freeze and execution records;
- later class-transfer states;
- and other anchor-owned bookkeeping or scientific dispositions that do not require their own theorem conclusion.

Their absence from a Lean result datatype does not remove them from the protocol.

The machine surface is selective by design.

Its purpose is to formalize claim transitions and firewalls where formal enforcement is useful—not to reproduce the entire scientific taxonomy for symmetry.

This boundary remains acceptable only while an omitted state cannot create a machine-significant shortcut.

If a later audit shows that an omitted scientific distinction must become machine-visible in order to preserve a claim boundary, that finding creates **adapter pressure**.

Until then, the scientific anchor remains authoritative over those states.

Interpretation Formula

The governing interpretation rule is:

SCIENTIFICALLY VALID, FAITHFULLY POPULATED RECORD
+
RELEVANT MACHINE CERTIFICATE
→
ANCHOR-LICENSED SCIENTIFIC RESULT

Never:

MACHINE CERTIFICATE → EMPIRICAL TRUTH

Firewalls and Forbidden Shortcuts

The released adapter is designed to prevent or expose the following invalid moves.

1. NOT-LIVE treated as FAIL.

A condition that the registered route does not require is not a failed condition.

2. NOT-LIVE treated as PASS.

Lack of liveness does not constitute successful performance of the condition.

3. UNRESOLVED treated as FAIL.

Insufficient adjudication is not positive failure.

4. UNRESOLVED treated as PASS.

Insufficient adjudication is not positive success.

5. Downstream liveness bypassing a NOT-LIVE predecessor.

A later structural condition cannot remain live when its required predecessor has been declared not live.

6. A later failure overriding first-break prerequisite order.

The first live failure owns the first-break classification. A later failure does not replace it.

7. J_ROUTE FAIL treated as a named first break without class evidence.

Route failure is necessary for a named first break, but is not sufficient.

8. PASS treated as a sixth Failure Regime.

PASS is the non-failure control state, not an additional failure class.

9. A secondary scale replacing the primary confirmatory result after target access.

Secondary scales remain diagnostic unless prospectively registered as primary in a later execution.

10. Morphology or held-out probe evidence used to manufacture intrinsic class assignment.

Failure-Regime classification must remain upstream of the probe result used to test it.

11. SEARCH FAILURE treated as evidence against the branch prediction itself.

Failure to place the branch under pressure is not failure of the branch hypothesis.

12. PHYSICALIZATION, SOURCE, LIGHT, MEASUREMENT, or O₂ failure treated as foundational novelty.

Each failure remains at the narrowest layer actually contradicted.

13. U₁ EARNED treated as proof that five-regime closure survived.

Incremental predictive utility and foundational closure answer different questions.

14. Novel observable pressure promoted directly to SRW.

$R_u > \varepsilon_u$ opens novelty pressure, not a structural residual witness.

15. SRW entered without the full **RouteUnknownHandoff.**

The route must preserve:

**clean branch setup + adequate search + valid entry / registration + primary scale +
E + route failure + no known first break**

before SRW entry is licensed.

16. SRW promoted directly to FR₆.

The replication, state-completeness, suspension, candidate, and newness burdens cannot be skipped.

17. An unreplicated SRW suspending closure.

One structural residual witness is insufficient for foundational suspension.

18. An incomplete state-completeness ladder suspending closure.

Closure suspension requires the pre-frozen ordinary-state ladder to have been exhausted.

19. Richer-state resolution ignored in order to preserve novelty.

If a richer admitted ordinary state resolves the witness, the correct result is state-incompleteness resolution.

20. Candidate rejection treated automatically as witness resolution.

A failed K^* candidate can leave the underlying residual completely unresolved.

21. Known-capacity reduction treated as closure restoration without scientific confirmation that the witness was actually resolved.

Reduction alone does not restore closure.

The residual itself must be accounted for.

22. Intervention unavailability used as a post-hoc waiver.

Intervention availability must be adjudicated under the predeclared rule, not invoked opportunistically after the result.

23. Reduction to a known capacity ignored because the candidate has a new name.

Terminological novelty does not establish structural novelty.

24. Functional equivalence ignored because the parameterization differs.

A changed mathematical or descriptive representation does not create a new structural job if the candidate is functionally equivalent to known structure.

25. A diagnostic **ResidualMilestoneCertificate used to under-report an active stronger final residual state.**

Milestones preserve history.

They do not override the strongest supported final report.

26. CLOSURE SUSPENDED reported as closure survived.

Suspension is an active restriction on a survival claim.

27. reopenedAtTestedScope reported as closure survived.

Reopening is not survival.

28. An unresolved or untested branch assigned an invented first-break class instead of none.

Missing class information must remain missing.

29. An unresolved or untested branch assigned an invented closure status instead of none.

No closure result is not equivalent to unchanged.

30. A closure-survival claim made where the foundational closure test was never actually placed under pressure.

Ordinary branch testing is not automatically foundational testing.

31. A provisional or unfinalized closure status used in a final program survival claim.

Program closure reporting requires finalized status.

32. A missing registered branch silently dropped from the final program record.

Complete accounting includes adverse, unresolved, and untested registered branches.

33. Successful branches selected after target access to manufacture a cross-regime U_1 claim.

The support set is frozen prospectively.

Later success cannot rewrite it.

34. Incompatible branches averaged into one global score.

Branch validity does not imply compatibility for one aggregate claim.

35. U_1 aggregation used to override foundational closure status.

Predictive utility cannot erase a suspended or reopened foundational state.

36. Lean elaboration success treated as empirical validation.

A clean compile establishes that the encoded formal object elaborates and the proofs check.

It does not establish that the scientific premises are true of nature.

Firewall Compression

The entire firewall set can be compressed into five operator rules.

Preserve scientific state distinctions exactly.

PASS, FAIL, NOT-LIVE, UNRESOLVED, OPEN, VIOLATED, and DISCHARGED perform different jobs.

Preserve causal and adjudicative order.

Do not let downstream observations rewrite upstream registration, classification, or freeze state.

Escalate only through the licensed handoff.

Ordinary failure does not become structural novelty, novelty does not become SRW, SRW does not become suspension, and suspension does not become FR_6 without the intervening burdens.

Preserve negative and unresolved evidence.

Do not invent classes, closure states, branch success, or scientific completeness where the execution did not earn them.

Keep machine consequence below scientific authority.

Lean checks the conditional structure supplied to it.

It does not supply the empirical premises.

Troubleshooting and Failed-Proof Interpretation

Troubleshooting Rule

When a requested proof or certificate is unavailable, diagnose the **lowest layer that explains the failure**.

Do not change the science merely because a Lean goal is inconvenient.

Do not strengthen Lean assumptions merely because the desired scientific conclusion is attractive.

The default diagnostic order is:

1. check the exact package and load identity;
2. check the record identity and scope;
3. check required burdens and their dispositions;
4. check scientific predecessor states;
5. check whether the requested theorem is actually the correct machine route for the scientific result;
6. only then treat the issue as a formalization defect.

The general rule is:

A blocked proof is evidence of a blocked proof until the scientific or semantic reason for that blockage has been established.

It is not automatically a failed experiment.

Open Burden

Meaning

A required scientific adjudication or setup burden remains unresolved.

Machine Effect

Ready predicates such as:

- **EntryReady;**

- **BranchSetupReady;**
- **ResidualEntryReady;**
- or **ProgramClaimFreezeReady**

cannot be constructed where a required burden must be discharged.

Operator Action

Return to the anchor-owned protocol and execution record.

Preserve the open state.

Do not manufacture:

Disposition.discharged

inside Lean.

Schematic Example

The O₂ comparator was not fully frozen before target access.

The branch may still contain scientifically interesting physical output.

But the machine route requiring:

ordinaryComparatorFrozen

is unavailable for confirmatory U₁ adjudication.

The correct result is not to pretend the freeze existed.

Violated Entry or Setup Burden

Meaning

A required condition for the requested confirmatory route positively failed.

Machine Effect

The corresponding ready predicate is unavailable.

Operator Action

Report at the narrowest scientific layer owned by the anchor.

Do not escalate downstream.

Do not recode a violated burden as open merely to avoid the consequence.

Schematic Example

The held-out probe was inspected before classification freeze.

Therefore:

heldOutProbeExcluded

or:

targetHoldoutIntact

cannot honestly be discharged.

The confirmatory machine route is unavailable.

A later exploratory analysis may still be scientifically useful.

It does not retroactively repair the original confirmatory freeze.

Malformed or Inconsistent Record

Meaning

The populated record violates:

- registration consistency;
- prerequisite-closed liveness;
- scope binding;
- or another required scientific relation.

Machine Effect

Named classification or later certificates are blocked.

Operator Action

Treat the record as inadmissible for that confirmatory route.

If a legitimate repair exists, perform it under the anchor's amendment rules.

Otherwise preserve the invalidity.

Schematic Example

Suppose:

P = NOT-LIVE

while:

B = PASS

The correct response is not to choose Shear or Drift.

The liveness declaration itself is inconsistent.

Requested Class Does Not Match the Prefix

Meaning

A known route state exists, but the requested named class is not supported by its predecessor and break evidence.

Machine Effect

FirstBreakEvidence for the requested class is unavailable.

Operator Action

Ask which class, if any, the actual record licenses.

Do not let later morphology determine the class.

Schematic Example

Suppose:

P₀ = PASS

P = FAIL

B = FAIL

The first live break is **Shear**.

The later Boundary failure does not promote the route to Drift.

Known First Break Found During Residual Review

Meaning

The known five-regime map accounts for the route failure.

Machine Effect

SRWCandidate is blocked because **RouteFailureWithoutKnownBreak** does not hold.

Operator Action

Report the known first-break result.

The unfamiliar probe appearance may remain scientifically interesting.

It is not foundational novelty on that route.

State-Incompleteness Resolved

Meaning

A richer pre-frozen ordinary state absorbs the apparent residual into known physics or a known first break.

Machine Effect

StateIncompletenessResolved is available.

MaximalStateStructuralResidual and **ClosureSuspended** are blocked.

Operator Action

Report the resolution.

Do not ignore the richer-state explanation because novelty was the hoped-for result.

Closure Suspended but FR₆ Not Earned

Meaning

A replicated maximal-state structural residual persists, but no stronger restoration or FR₆-earned state has been established.

This includes:

- no positive candidate yet exists;
- a candidate remains unresolved;
- or a candidate has failed while the residual witness itself remains unresolved.

Machine Effect

The final **ResidualReportingCertificate** may return:

closureSuspended

and:

ClosureSurvivalReportAllowed = False

Operator Action

Preserve:

CLOSURE SUSPENDED

Do not report:

closure survived

and do not report:

FR₆ EARNED

Suspension is a legitimate terminal public state while further scientific work remains outstanding.

Candidate Fails: Restore or Remain Suspended

Meaning

candidateNewnessState is violated.

But candidate failure and residual-witness resolution remain separate adjudications.

Restoration Path

If:

- the candidate reduces to known structure or functional equivalence;
- and **knownCapacityReductionResolvesWitness = True**;

then the final machine result is:

KnownCapacityReductionResolved

and closure becomes:

restoredAtTestedScope

Continued-Suspension Path

If the candidate fails but the witness remains unresolved, the final machine result remains:

closureSuspended

Operator Action

Do not infer restoration merely from non-newness.

Populate the witness-resolution adjudication from the scientific record.

Program Aggregation Blocked

Meaning

Branch-local results remain reportable, but the requested program claim lacks one or more required aggregation or closure-reporting burdens.

Common U₁ Causes

- a registered branch record is missing;
- a support branch was not actually tested;
- a support branch did not earn U₁;
- the program object does not match;
- the claim content does not match;
- support branches are incompatible;
- the required scale relation is not admitted;
- **firstBreakClass = none** where a cross-regime distinction is required;
- or no genuinely cross-regime pair exists.

Additional Closure-Survival Causes

- the closure test was not actually placed under pressure;
- the closure status is not finalized;
- **closureStatusAt = none**;
- closure is suspended;
- or closure is **reopenedAtTestedScope**.

Operator Action

Preserve the branch-level results.

Report the aggregate claim as unavailable.

Do not substitute a successful branch after target access.

Do not invent missing class or closure values.

U₁ License Present, Closure License Absent

Meaning

The program can satisfy the frozen aggregate predictive-utility claim while at least one relevant branch has suspended or reopened foundational closure.

Machine Effect

CrossRegimeUtilityClaimLicensed may exist while:

ProgramClosureSurvivalLicensed

remains blocked.

Operator Action

Report the two results separately.

Do not let utility success erase structural pressure.

Scientific State Has No Dedicated Lean Constructor

Meaning

The scientific protocol requires a record state that the adapter represents as:

- a premise;
- a blocker;
- an external execution-schema value;

- or an anchor-owned scientific result

rather than as a dedicated Lean conclusion.

Examples include:

- DCR provenance;
- known-mixture states;
- purity states;
- coalescence states;
- purity or novelty discrimination underpowered states;
- some calibration and freeze records;
- later class-transfer states;
- and transfer-underpowered states.

Operator Action

Report the state exactly as the anchor requires.

Do not invent a Lean constructor inside the companion.

Do not claim the scientific state is unsupported merely because the adapter does not enumerate it.

Open adapter-pressure adjudication only if the missing machine representation prevents enforcement of a **machine-significant claim boundary**.

Do not enlarge the theorem surface merely for taxonomy symmetry.

Lean Elaboration or Proof Error

Default Disposition

MACHINE ONLY until demonstrated otherwise.

Meaning

The Lean source may contain:

- a namespace error;

- syntax defect;
- proof defect;
- import or load-order problem;
- or type-engineering issue.

Operator Action

Repair the formalization first.

Reopen scientific semantics only if the repair attempt demonstrates that a distinction required by the anchor cannot be represented faithfully.

A compiler message is not scientific evidence.

Semantic Mismatch Between Files

Meaning

The scientific anchor, adapter, companion, or kernel appears to assign different meanings or claim ceilings to the same public concept.

Operator Action

Stop ordinary execution and classify the pressure before repair.

Possible classifications are:

COMPANION GAP

The explanation is wrong or incomplete.

ANCHOR GAP

The scientific meaning is genuinely absent or contradictory.

ADAPTER GAP

The machine route omits, weakens, or overstates an anchor burden.

KERNEL PRESSURE

A required reusable domain-neutral capacity cannot be represented by the current Universal Kernel.

Repair at the **lowest authority level sufficient to answer the demonstrated pressure.**

Troubleshooting Exit Rule

Never resolve a machine blockage by silently strengthening the premises.

The scientific record must remain the source of machine facts.

If the requested claim cannot be proved from the honest record, one of four things follows:

- a narrower result is correct;
- the state remains unresolved;
- the machine route itself requires repair;
- or upstream semantics genuinely require reconsideration.

The desired conclusion does not decide which one applies.

Release and Execution Record

Two Kinds of Identity

The package preserves two distinct identity questions.

Semantic Compatibility

Does the scientific anchor, adapter, Universal Kernel, and companion agree in machine-relevant meaning?

Exact Release / Execution Binding

Which concrete files were packaged or used in one scientific execution?

These jobs must remain separate.

Stable semantic references prevent harmless version chasing.

Exact manifests preserve reproducibility.

Stable Scientific-Anchor Reference

The adapter and companion identify the scientific authority by the stable title:

Five Failure Regimes in Dark-Matter Light Probes: An Open-Source Pre-Result Demonstration Protocol

and by the rule:

Use the governing accepted edition supplied with the release package.

A harmless scientific-anchor version bump does not by itself require a new adapter or companion build.

Exact Release Binding Belongs in the Manifest

A release manifest should identify, at minimum:

- the exact scientific-anchor filename or release identity;
- the exact Dark-Matter Failure-Regime Adapter / Contract filename;
- the exact Structural Flow Universal Kernel filename;
- the exact companion filename;
- cryptographic hashes where practical;
- DOI or repository release identity where applicable;
- Lean or compiler environment where practical;
- and the package date or release identifier where useful.

A concrete scientific execution should additionally bind the experiment-owned execution identifiers required by the anchor, including:

- H_ROOT;
- target records;
- freeze records;
- and other execution identities required by the scientific protocol.

Package Manifest Identity Is Not H_ROOT

These two identities perform different jobs.

Package Manifest

The package manifest identifies which public artifacts belong together.

H_ROOT

H_ROOT identifies the experiment's pre-target freeze bundle for one concrete scientific execution.

Do not conflate them.

A package may be immutably released before any scientific execution exists.

A later scientific execution may then use that package while producing its own H_ROOT.

Machine Load Relation

The public machine files remain separate:

1. **Structural Flow Universal Kernel**
2. **Dark-Matter Failure-Regime Adapter / Contract**

For the tested browser concatenation workflow, the compatible Universal Kernel is loaded first and the adapter follows beneath it.

A packaged implementation may use an equivalent valid import or load relation.

The release manifest should identify the exact kernel / adapter pair that was machine-checked together.

The companion is **not imported by Lean**.

It is the human execution and interpretation manual for the packaged machine relation.

Compatibility Across Later Anchor Editions

Compatibility with later anchor editions follows the semantic—not numerical—rule defined under Package Identity and Compatibility above. Copyediting, citation correction, typography, explanatory expansion, and other non-machine-significant changes do not by themselves require a new adapter. If a later anchor changes a machine-relevant condition, result meaning, prerequisite or liveness relation, burden ownership, routing rule, residual/reopening gate, or program license/blocker, stable-title compatibility ends. Run a new compatibility audit and produce a new adapter build where required.

Companion Identity

The companion should normally iterate when:

- a public machine identifier changes;
- execution guidance changes materially;
- interpretation guidance changes materially;
- an authority boundary requires clarification;
- a new public machine route must be explained;
- or a cross-file audit earns a repair.

The companion need not chase harmless anchor-version bumps that preserve its semantic instructions.

Internal Scaffolding

The machine-integration spine and companion-production spine are **internal production artifacts by default**.

They should not be presented as governing scientific or machine authorities in the public release unless a separate publication decision explicitly says otherwise.

The public authority constellation remains:

scientific anchor

+ adapter / contract

+ Universal Kernel

+ specialist companion

Release-Readiness Identity Check

Before package handoff, confirm:

- the anchor's stable title is consistent everywhere;
- exact release identities appear in the manifest rather than being hard-wired into stable semantic prose;
- adapter and kernel compile as the exact packaged pair;
- companion public identifiers match the production adapter;
- no stale candidate or development wording implies a different authority state;
- and no internal production spine is accidentally presented as governing authority.

Execution Reporting Rule

A scientific result should identify both:

the anchor-defined execution and freeze identity supporting the scientific adjudication

and:

the exact public package release used to interpret and machine-check that adjudication

This permits later package maintenance without retroactively changing the identity of an earlier scientific execution.

Closing Operator Rules

The package can be operated faithfully by preserving a small number of discipline rules.

Physics owns the physical premises.

The machine layer does not decide what happened in the dark-matter system.

The scientific anchor owns adjudication and claim meaning.

The adapter does not rewrite the experiment.

Populate; do not reinterpret.

Every machine state must come from an identified scientific source.

Preserve OPEN, UNRESOLVED, NOT-LIVE, and none.

Incomplete information is not a defect to be filled with a favorable value.

Preserve the one-way experimental chain.

Later evidence cannot rewrite earlier registration or freeze state.

Use the strongest supported final result.

Do not hide suspension, restoration, or reopening behind an earlier diagnostic milestone.

Keep U_1 and foundational closure separate.

Predictive utility does not answer the foundational completeness question.

Repair at the lowest sufficient authority level.

Compiler problems begin as machine problems.

Explanation problems begin as companion problems.

Scientific-semantic problems belong to the anchor.

Only demonstrated reusable domain-neutral insufficiency creates kernel pressure.

Machine-clean is not physics-confirmed.

AI Use Disclosure

Artificial intelligence tools were used during the development of this work for research assistance, adversarial review, conceptual organization, drafting support, editing, consistency checking, and formalization support. The author directed the research program, determined the package architecture and authority boundaries, evaluated the scientific-to-machine mappings, and takes responsibility for the final content.